Paris Street Combat

Assignment 3: MVP Implementation & Demo | 10 October 2026

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1. Plan vs. Reality

Following the completed and approved Assignment 2, we built the core technical slice: animation, collision, navigation and independent NPC decisions working together in the Paris environment. The playable mission starts across the river: cross the bridge with two allies, defeat three German guards and capture G1. A highlighted safe zone then offers a consent-based checkpoint save. Loading restores progress and resources; player death supports mission restart.

Graphics pillar	Implemented and verified in the MVP
Animation	Player walk/run/slow movement, crouch, crawl, jump, aim, fire and reload; NPC movement/combat and death. Weapon events conserve ammunition, and saved corpses restore directly to the terminal death pose.
Collision Detection	Character/environment collision blocks movement. Camera-to-aim and muzzle-to-aim traces resolve cover, soldier obstruction and damage. Health and ammunition displays read the same authoritative gameplay state.
Pathfinding and Navigation	UE NavMesh/MoveTo drives real character travel. Map surveys and formal-character tests established the mission route and bridge crossing; allies follow/regroup at distinct goals, with bounded replanning and blocked-path recovery.
NPC AI / Behavior Trees	Shared behavior definitions use separate controllers and Blackboards. Faction/perception drives engagement, remembered targets, timed search and return to duty; squad goals, reservations and death cleanup coordinate allies.

Result and scope changes. All four pillars are complete and accepted for the current MVP, supported by retained tests and manual review. The midterm ends at G1; city-center occupation and later stages are deferred. Checkpoints moved from Should to implemented. We use UE navigation rather than the optional custom tactical A*. The initial roster remains one player, two allies and three guards.

2. Technical Verification and Performance

Implementation uses UE 5.8.2, existing Animation Blueprints, native C++ gameplay and presentation modules, character collision/traces, NavMesh/MoveTo and Behavior Trees. Retained stage-specific evidence includes the October 2 combat/HUD regression (15 cases, 62 assertions), October 3 player-action regression (16 scenarios, 66 assertions), October 7 NPC acceptance B00-B05, and subsequent G1/save/load/restart and manual checks. Validation spans dated builds with the scopes recorded in each test receipt. A teammate also confirmed the shared game runs and is playable on a second computer.

Target: 60 FPS at 1920 x 1080 on i9-12900F / RTX 3080 / 32 GB Windows PC. Measured at High, ray tracing/VSync off, uncapped, with the existing roster. Ready baselines use offscreen rendering.

10 October measurement	Mean FPS	Mean / p95 frame time
Normal audio V2, stationary Ready 3 runs; 30 s warmup, then 91-93 s each	101.04 / 101.16 / 102.14	9.90 / 11.04 ms 9.89 / 11.14 ms 9.79 / 10.78 ms
Separate V6 input-helper run Crossing to Won, approximately 60.26 s	94.08	10.63 / 12.20 ms

Ready measurements test rendering with NPC brains gated off. The helper run includes telemetry overhead and approximate phase alignment. Means exceed the target in these measured windows, not every frame or arbitrary NPC load. Its whole capture retains an 8.47 s loading hitch. The approved video shows live UE timing/RAM/VRAM counters during the existing-roster workload, including roughly 8 s load/restart stalls and a brief texture-budget warning; diagnostics, input assistance and OBS add overhead. This reports local limits without claiming an isolated hardware cause or large-population capacity.

3. AI Utility

First, we tried GPT-assisted detailed modeling. The team had no professional 3D modeling experience. Repeated AI-designed/executed character trials, including a 38-iteration experiment, did not produce assets meeting this project's production needs. We stopped that route and moved to existing assets.

We then purchased existing licensed assets. This provided substantially better base models, but they were not ready to integrate unchanged. Hands, weapons, character poses and animations needed fitting. The initial set lacked a suitable German rifle and corresponding effects/audio. We addressed these gaps incrementally through compatible assets, bounded adaptation, attachments, event timing and in-game verification. Model processing and adaptation took approximately **10 working days (team estimate)**, longer than expected. The current build integrates armed NPCs, muzzle effects and gameplay sound.

Value and cost. ChatGPT/Codex helped turn requirements into code, diagnose engine/import issues, build validation tools and document failures. ImageGen helped with concept and HUD references; the playable HUD implements the approved design. Detailed runtime models primarily come from purchased assets. AI was useful for implementation and diagnostics, while modeling retries and asset integration consumed more time than planned. AI narration is a demo-production aid. The proposal did not commit to runtime LLM/API calls: NPC decision-making runs locally through UE Behavior Trees.

4. Roadmap to Final

Remaining work	Next implementation and verification
Reload sleeve obstruction	Correct the sleeve that still occludes the first-person camera during reload. Preserve accepted models, finger poses and weapon/ammunition transactions; verify the complete action in motion.
HUD refinement	Improve layout, visual details, feedback and readability across resolutions while retaining the working health, ammunition, minimap and checkpoint UI.
Richer gameplay	Extend the G1 loop with reviewed mission stages, encounter pacing, route choices and squad cooperation. Decide specific additions through design review.
More NPCs and stress tests	Extend multiple-squad roles, target assignment, perception/attack, congestion recovery and mission lifecycle. Test increasing populations for behavior correctness, frame times, memory and stalls. Current small-roster acceptance does not establish this future capacity.

5. Deliverables and Access

Playable Windows build: selected audio V2, **Paris-G1-Audio-V2-20261009.zip** (8.48 GB), with executable, cooked assets and runtime prerequisites. Use the separately supplied private course attachment **SFTPAccess_20261010**: run **DOWNLOAD_GAME.cmd**, wait for download and SHA-256 verification, extract the entire game archive, then run **PLAY_G1_REVISION.cmd**. Windows OpenSSH is required for download; allow at least 18 GB free disk space. Credentials stay outside this report and public Git.

Private SFTP is the selected alternative to the specified Drive/itch.io channel; instructor approval of that channel remains to be confirmed. The hosting PC must remain online. Course attachment upload and independent reviewer download are separate from the teammate playability check.

Real-time demo (2:38): https://youtu.be/SktbFHNYP54
English embedded captions, AI voiceover and live performance display; includes checkpoint restore and enemy-caused player death/restart. Normal-speed input assistance, separate takes and cuts are disclosed. The 30 FPS video export rate is separate from game FPS.

Public source: https://github.com/hzgyp/CS549-Paris-Street-Combat
README links the build/restoration workflow; the current-build guide identifies the selected 42-file source and licensed dependencies. Public source does not include commercial asset bytes; restore authorized assets before rebuilding.